P0 REVIEW ONLY | Power links JP1–JP8 OPEN | No OLED footprint | Module harness order defined here; verify module end separately.

REGULATED 5V INPUT
J1
5V_IN 1
GND 2

Q1
AO3401A
GND 1
5V_IN 2
5V_PROT 3

C1
100u 10V
5V_PROT
GND

C2
1u X7R 10V
5V_SYS
GND

U1
MCP1700-3302E/TO
GND 1
5V_SYS 2
VI
GND
VO
3V3_AUX 3

C3
1u X7R 10V
3V3_AUX
GND

C4
10u 10V
3V3_AUX
GND

MASTER POWER – OPEN
JP1
5V_PROT 1
5V_SYS 2

C5
100n 10V
3V3_BP
GND

PWR_FLAG
#FLG01
5V_SYS

POWER: 5V source –> reverse polarity FET –> Blue Pill onboard regulator + independent AUX regulator. NEVER JOIN 3V3_BP AND 3V3_AUX.

Blue Pill nominal rows: 20 pins, pitch 2.54mm, row spacing 15.24mm. J2/J3 pin 1 at SWD end. Actual board measurement required.

PWR_FLAG
#FLG02
GND

BLUE PILL H1 – NOMINAL
J2
REED_PA1 6
SD_CS_PA4 9
SD_SCK_PA5 10
SD_MISO_PA6 11
SD_MOSI_PA7 12
BUZZER_PB0 13
GREEN_PB1 14
BLUE_PB10 15
RED_PB11 16
NRST 17
3V3_BP 18
GND 19
GND 20

BLUE PILL H2 – NOMINAL
J3
3V3_BP 1
GND 2
5V_SYS 3
I2C_SDA_PB7 6
I2C_SCL_PB6 7
DHT_PB5 8
GPS_TX_PA10 14
GPS_RX_PA9 15
BUTTON2_PB14 18
BUTTON1_PB13 19
SW420_PB12 20

MPU HARNESS – VERIFY
J4
MPU_VCC 1
GND 2
I2C_SCL_PB6 3
I2C_SDA_PB7 4
MPU_VCC 5

MPU POWER – OPEN
JP2
3V3_AUX 1
MPU_VCC 2

C6
100n 10V
MPU_VCC
GND

RTC HARNESS – VERIFY
J5
RTC_VCC 1
GND 2
I2C_SCL_PB6 3
I2C_SDA_PB7 4

RTC POWER – OPEN
JP3
3V3_AUX 1
RTC_VCC 2

C7
100n 10V
RTC_VCC
GND

R1
4k7 DNP
3V3_AUX
I2C_SCL_PB6

I2C: MPU AD0 high, address 0x69; RTC 0x68. Pullups R1/R2 DNP until module pullups audited. Failed MPU/SD tests remain unresolved.

R2
4k7 DNP
3V3_AUX
I2C_SDA_PB7

DHT11 HARNESS – VERIFY
J6
DHT_VCC 1
GND 2
DHT_PB5 3

DHT POWER – OPEN
JP4
3V3_AUX 1
DHT_VCC 2

R3
4k7 DNP
DHT_VCC
DHT_PB5

C8
100n 10V
DHT_VCC
GND

GPS HARNESS – VERIFY
J7
GPS_VCC 1
GND 2
GPS_TX_PA10 3
GPS_RX_PA9 4

GPS 5V CANDIDATE – OPEN
JP5
5V_SYS 1
GPS_VCC 2

C9
100n 10V
GPS_VCC
GND

SD HARNESS – VERIFY
J8
SD_VCC 1
GND 2
SD_CS_PA4 3
SD_SCK_PA5 4
SD_MISO_PA6 5
SD_MOSI_PA7 6

Sensor supply links are isolated by default. SD/GPS candidate 5V must NOT be connected until breakout regulators / logic are identified.

SD 5V CANDIDATE – OPEN
JP6
5V_SYS 1
SD_VCC 2

C10
100n 10V
SD_VCC
GND

C11
100u 10V
SD_VCC
GND

R4
10k
3V3_BP
SD_CS_PA4

SW420 HARNESS – VERIFY
J9
SW420_VCC 1
GND 2
SW420_PB12 3

SW420 POWER – OPEN
JP7
3V3_AUX 1
SW420_VCC 2

C12
100n 10V
SW420_VCC
GND

REED TO GND
J10
REED_PA1 1
GND 2

BUTTON 1 TO GND
J11
BUTTON1_PB13 1
GND 2

PB12 SW420 is proposed, not hardware-tested. Older firmware PIN_SW420=PB15; synchronize before bring-up.

BUTTON 2 TO GND
J12
BUTTON2_PB14 1
GND 2

R5
10k
3V3_BP
REED_PA1

R6
10k
3V3_BP
BUTTON1_PB13

R7
10k
3V3_BP
BUTTON2_PB14

R8
330
GREEN_PB1
LED_GREEN_A

R9
330
BLUE_PB10
LED_BLUE_A

R10
330
RED_PB11
LED_RED_A

GREEN
D1
GND
LED_GREEN_A

BLUE
D2
GND
LED_BLUE_A

LEDs have 330-ohm resistors. Buzzer has transistor + flyback diode; candidate load <=20mA / 5V pending actual buzzer specifications.

RED
D3
GND
LED_RED_A

R11
680
BUZZER_PB0
BUZZER_BASE

R12
100k
BUZZER_BASE
GND

Q2
P2N2222AG
BUZZER_BASE 2
BUZZER_LOW
3V3
GND

BUZZER HARNESS
J13
BUZZER_VCC 1
BUZZER_LOW 2

BUZZER 5V – OPEN
JP8
5V_SYS 1
BUZZER_VCC 2

1N4007
D4
BUZZER_VCC
BUZZER_LOW

TO BLUE PILL SWD CABLE
J14
SWDIO_PA13 1
SWCLK_PA14 2
GND 3

STLINK SIGNALS ONLY
J15
SWDIO_PA13 1
SWCLK_PA14 2
GND 3
NRST 4

SWD requires J14 cable to the separate Blue Pill SWD header. J15 has NO power output. J16 is unpopulated VTref SENSE ONLY.

VTREF SENSE ONLY
J16
3V3_BP 1
GND 2

GND
TP1
GND

5V_PROT
TP2
5V_PROT

3V3_AUX
TP3
3V3_AUX

3V3_BP
TP4
3V3_BP

M3 3.2mm
H1

M3 3.2mm
H2

M3 3.2mm
H3

M3 3.2mm
H4

Blue Pill carrier – module identities and mechanics pending
Sheet: /
File: SmartLogistics.kicad_sch
Title: Smart Logistics – PROVISIONAL / DO NOT FABRICATE
Size: A2 | Date: 2026-10-16 | Rev: P0
KiCad E.D.A. 10.0.7 | Id: 1/1